BRAC University
MAT-215
Exercise Sheet #1

Part A

1. Perform each of the indicated operations:

(i) $(i-2)\{2(1+i)-3(i-1)\}$ (ii) $\dfrac{(2+i)(3-2i)(1-i)}{(1-i)^2}$ (iii) $(2i-1)^2\left\{\dfrac{4}{1-i}+\dfrac{2-i}{1+i}\right\}$

(iv) $3\left(\dfrac{1+i}{1-i}\right)^2-2\left(\dfrac{1-i}{1+i}\right)^3$ (v) $\dfrac{3i^{10}-i^{19}}{2i-1}$ (vi) $\dfrac{i^4-i^9+i^{16}}{2-i^5+i^{10}-i^{15}}$

2. Show that (i) $(5+3i)+\{(-1+2i)+(7-5i)\}$ and (ii) $\{(5+3i)+(-1+2i)\}+(7-5i)$ illustrate the associative law of addition.

3. If $z_1=1-i,\ z_2=-2+4i,\ z_3=\sqrt{3}-2i$, evaluate each of the following:

(i) $|2z_2-3z_1|^2$ (ii) $\left|\dfrac{z_1+z_2+1}{z_1-z_2+i}\right|$ (iii) $\overline{(z_2+z_3)(z_1-z_3)}$ (iv) $\mathrm{Re}\{2z_1^3+3z_2^3-5z_3^2\}$

(v) $\mathrm{Im}\left\{\dfrac{z_1z_2}{z_3}\right\}$ (vi) $z_1^2+2z_1-3$ (vii) $|z_1\overline{z_2}+z_2\overline{z_1}|$ (viii) $\dfrac{1}{2}\left(\dfrac{z_3}{\overline{z_3}}+\dfrac{\overline{z_3}}{z_3}\right)$

(ix) $(z_3-\overline{z_3})^5$

4. Express each of the following complex number in polar form and show them graphically.

(i) $2+2\sqrt{3}\,i$ (ii) $2\sqrt{2}+2\sqrt{2}\,i$ (iii) $-2\sqrt{3}-2i$ (iv) $-1+\sqrt{3}\,i$ (v) $-\sqrt{6}-\sqrt{2}i$

5. Prove that: (i) $\overline{z_1+z_2}=\overline{z_1}+\overline{z_2}$ (ii) $|z_1z_2|=|z_1||z_2|$ (iii) $\overline{z_1-z_2}=\overline{z_1}-\overline{z_2}$
(iv) $|z|\geq \mathrm{Re}(z)$.

6. (i) State **De Moivre's Theorem**
(ii) Apply **De Moivre's Theorem** to find $(1+i)^{20}$

7. Evaluate each of the following by **De Moivre's Theorem:**

(i) $\dfrac{(8cis40^0)^3}{(2cis60^0)^4}$ (ii) $\dfrac{(3e^{\frac{\pi i}{6}})(2e^{\frac{-5\pi i}{4}})(6e^{\frac{5\pi i}{3}})}{(4e^{\frac{2\pi i}{3}})^2}$ (iii) $(5cis20^0)(3cis40^0)$ (iv) $(2cis50^0)^6$

8. Find all the roots of the following equations.

(i) $(-1+i)^{\frac{1}{3}}$ (ii) $z^5=-4+4i$ (iii) $z^4=-16i$ (iv) $z^6=64$ (v) $z^4+z^2+1=0$.

$(vi)\ (-4+4i)^{1/5}$ $(vii)\ (-2\sqrt{3}-2i)^{1/4}$

Part B

1. Perform the indicated operations analytically and graphically.
   (a) $(2+3i)+(4-5i)$ (b) $(7+i)-(4-2i)$
2. Describe geometrically the set of points z satisfying the following conditions:
   (a) $Re(z)>1$
   (b) $|2z+3|>4$
   (c) $Re(\frac{1}{z})>1$
   (d) $1<|z-2i|<2$
   (e) $|z+1-i|\leq|z-1+i|$
   (f) $Re(z)\geq 0$
   (g) $|z-4|\geq|z|$
   (h) $|z-2|\leq|z+2|$
   (i) $Re(1/z)\leq 1/2$
   (j) $\pi/2<\arg z<3\pi/2,\ |z|>2$

3. Using the properties of conjugate and modulus, show that:
   I. $\overline{\bar{z}+3i}=z-3i$
   II. $|(2\bar{z}+5)(\sqrt{2}-i)|=\sqrt{3}\ |2z+5|$
   III. $|2z+3\bar{z}|\leq 4|Re(z)|+|z|$ {Hint: Use triangular inequality $|z_1\pm z_2|\leq|z_1|+|z_2|$ or $|z_1\pm z_2|\geq|z_1|-|z_2|$}.

4. Find the modulus and argument of the following complex numbers:
   (i) $\frac{2-i}{2+i}$
   (ii) $\frac{\sqrt{3}+i}{\sqrt{3}-i}$
   (iii) $\left(\frac{1+\sqrt{3}i}{1-\sqrt{3}i}\right)^2$

5. Prove that |z-i| = |z+i| represents a straight line.
6. Prove that $|z+2i|+|z-2i|=6$ represents an ellipse.
7. Find an equation of a circle center at (2,3) with radius 3.
8. Sketch the region in xy- plane represented by the following set of points:
   $Re(\bar{z}-1)=2$
   $Im(z^2)=4$

$|\frac{2z-3}{2z+3}|=1$

$Re(z) + Im(z) = 0$

BRAC University
MAT-215
Exercise Sheet # 2

1. Evaluate the following Limits:

(i) $\lim\limits_{z\to 1+i} \dfrac{z^2 - z + 1 - i}{z^2 - 2z + 2}$ (ii) $\lim\limits_{z\to 1+i} \left\{\dfrac{z - 1 - i}{z^2 - 2z + 2}\right\}^2$ (iii) $\lim\limits_{z\to i} \dfrac{z^2 + 1}{z^6 + 1}$.

2. Use the definition of neighborhood to solve the following problems:

(i) Show that $\lim\limits_{z\to i} 2z = 2i$

(ii) Show that if $f(z) = \frac{i\bar{z}}{2}$ then $\lim\limits_{z\to 1} f(z) = \frac{i}{2}$.

3. If $f(z) = \dfrac{2z - 1}{3z + 2}$, prove that $\lim\limits_{h\to 0} \dfrac{f(z_0 + h) - f(z_0)}{h} = \dfrac{7}{(3z_0 + 2)^2}$ provided $z_0 \neq -\dfrac{2}{3}$.

4. Let $f(z) = \dfrac{z^2 + 4}{z - 2i}$ if $z \neq 2i$, while $f(2i) = 3 + 4i$, is $f(z)$ *continuous at* $z = 2i$.

5. Find all points of discontinuity for the function $f(z) = \dfrac{2z - 3}{z^2 + 2z + 2}$.

6. Using the definitions, find the derivative of each function at the indicated points

(i) $f(z) = \dfrac{2z - i}{z + 2i}$ at $z = -i$

(ii) $f(z) = 3z^{-2}$ at $z = 1 + i$.

7. Evaluate the following Limits using L' Hospital's rule

(i) $\lim\limits_{z\to 2i} \dfrac{z^2 + 4}{2z^2 + (3 - 4i)z - 6i}$ (ii) $\lim\limits_{z\to 0} \dfrac{z - \sin z}{z^3}$ (iii) $\lim\limits_{z\to 0} \left(\dfrac{\sin z}{z}\right)^{\frac{1}{z^2}}$

8. Determine which of the following functions u are harmonic. For each harmonic function find the conjugate harmonic function v and express $u + iv$ as an analytic function of z

(i) $u = 3x^2 y + 2x^2 - y^3 - 2y^2$

(ii) $u = xe^x \cos y - ye^x \sin y$

(iii) $u = e^{-x}(x \sin y - y \cos y)$.

BRAC University
MAT-215
Practice Sheet # 3

1. Show that:

   $(i)\ Exp(2\pm 3\pi i) = -e^2 \quad (ii)\ Exp\left(\dfrac{2+\pi i}{4}\right) = \sqrt{\dfrac{e}{2}}(1+i) \quad (iii)\ Exp(z+\pi i) = -e^z.$

2. Find all values of z such that:

   $(i)\ e^z = -2 \quad (ii)\ e^z = 1+\sqrt{3}\,i \quad (iii)\ e^{2z-1} = 1 \quad (iv)\ e^z = -1.$

3. Prove that:

   $(i)\ \sin z = \sin x \cosh y + i \cos x \sinh y$

   $(ii)\ \cos z = \cos x \cosh y - i \sin x \sinh y$

   $(iii)\ \sin(z+2\pi) = \sin z$

   $(iv)\ \cos(z+2\pi) = \cos z.$

   $(v)\ \sin(z_1+z_2) = sinz_1 cosz_2 + cosz_2 sinz_2$

4. Prove that: $(i)\ \sinh z = \sinh x \cos y + i \cosh x \sin y$

   $(ii)\ \cosh z = \cosh x \cos y + i \sinh x \sin y.$

5. Show that: $(i)\ \sin^{-1} z = -i \ln[iz \pm (1-z^2)^{1/2}]$

   $(ii)\ \cos^{-1} z = -i \ln[z \pm i(1-z^2)^{1/2}].$

   $(iii) tanh^{-1}z = \frac{1}{2} ln\left(\frac{1+z}{1-z}\right).$

   $(iv) tan^{-1}z = \dfrac{1}{2i} ln\left(\dfrac{1+iz}{1-iz}\right)$

6. Solve the following equations: $(i)\ \cosh z = \dfrac{1}{2} \quad (ii)\ \sinh z = i.$

7. Show that: $(i)\ \ln\left(-1+\sqrt{3}\,i\right) = \ln 2 + 2\left(n+\dfrac{1}{3}\right)\pi i$

   $(ii)\ \ln(1-i) = \dfrac{1}{2}\ln 2 + \left(2n+\dfrac{7}{4}\right)\pi i$

   $(iii)\ \ln\left(i^{1/2}\right) = \left(n+\dfrac{1}{4}\right)\pi i.$